How do external reference prices influence online gift giving?

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Abstract: The authors examine the role of external reference prices in a unique form of gift giving behaviour – online gift registries. A hierarchical Bayesian analysis of fulfilment data from 555 online wedding registries reveals the probability of a gift fulfilment follows a bimodal distribution around the gift registry's average price. This is consistent with the hypothesis that online gift purchases are driven by the average price of the registry, which serves as the reference price, and two competing motivations among gift givers. These motivations are a desire for social benefits (e.g., to enhance the relationship) and a desire to limit monetary costs (e.g., to save money). The former motivation favours gifts with higher than average prices; whereas the latter favours those with lower than average prices. Because an average priced gift does not appeal to either segment, its fulfilment probability is relatively low. Finally, because gifts with extremely high or extremely low prices have less appeal for each segment, the fulfilment probability curve is bimodal.

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Keywords: external reference price; online gift registry; gift registry fulfilment; online retailing.

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1 Introduction

From the Trojan Horse in ancient Greece to the Statue of Liberty in modern America, for weddings, birthdays, anniversaries, and holidays, gift giving has been a truly universal phenomenon. Across vastly different social settings, gift giving is often an act of symbolism, large and small, of love, respect, and sometimes war. The economic importance of gift giving cannot be overestimated.

According to the Consumer Expenditure Survey conducted by the United States Bureau of Labor Statistics, Americans spent more than $137 billion on gifts in 2006, more than what they spent at fast food restaurants ($134 billion). Spurred by Sherry's (1983) theoretical framework, consumer researchers have extensively studied gift giving (e.g., Beatty et al., 1991; Dunn et al., 2008; Goodwin et al., 1990; Nguyen and Munch, 2011).

While the prior literature typically examines different generic forms of gift giving, our research provides a more focused study of online registry gift giving. Registry gift giving is a unique form of gift giving because, unlike other gift giving situations, the gift giver chooses from a clearly defined choice set prepared by the *recipient*. This approach controls for the impact of consideration sets, which are typically unobservable in other forms of gift giving (e.g., Roberts and Lattin, 1991), so it presents a clearer picture of various gift giving motivations. Moreover, for online registries, the average registry price can act as an *external reference price*.

An external reference price is often estimated from current retailer shelf prices (Hardie et al., 1993; Moon et al., 2006; Van Oest, 2013). Although a gift registry alleviates uncertainty about the recipient's taste, the individual items do not provide all the information that a gift giver needs. Thus, in addition to looking at individual items, a gift giver is likely to create an external reference price from the listed prices in the registry.

By combining insights from research on gift giving and external reference prices, our study develops the following research contributions. First, gift giving is examined in an important and unique context – online gift registries. Second, existing gift giving research is complemented by using data from seven retailer online gift registries. In contrast, prior research relies on either theoretical insights or experimentation (e.g., Camerer, 1988; Sherry, 1983). Third, unique insights arise by applying insights from external reference prices to gift giving. For example, a higher item price often leads to higher rather than lower fulfilment rates. Also, because an average registry price does not appeal to gift givers who are cost conscious or are trying to strengthen their social relationship, these items have some of the lowest fulfilment rates.

The remainder of the paper is organised as follows. We first review the research on gift giving motivations and reference prices and develop hypotheses on the role of external reference prices in gift fulfilment from online gift registries. This is followed by a discussion of the data, data analysis, and empirical results. The final section summarises the key findings, discusses limitations, and provides direction for future research.

2 Theory and hypotheses

Prior research on gift giving concludes that gift giving is primarily driven by two conflicting motivations (e.g., Beatty et al., 1991; Goodwin et al., 1990; Sherry, 1983; Wolfinbarger and Yale, 1993), In one case, a gift giver is trying to express their social identity and emotions for the gift recipient. The giver's behaviour is largely driven by the intensity of his or her feelings (affective or moral) towards the recipient. This type of gift giving is designed to strengthen the social relationship between the gift giver and the recipient.

Note, Dunn et al. (2008) report that spending money on others tends to increase happiness more than personal spending. Even so, most givers incorrectly believe that personal spending will increase happiness more than gifting. Thus, this secondary benefit from gift giving should not have a meaningful impact on the gift selection from an online wedding registry.

In the other case, however, gift giving takes on the nature of an economic exchange (Sherry, 1983). Here gifting costs are much more important. In other words, the gift giver

is, implicitly or not, performing a cost-benefit analysis. In this case, the giver thinks about costs first (either monetary or otherwise) and social benefits second.

Lowes et al. (1971) provide empirical support for these dual motivations, with 84% of gift givers being primarily motivated by either social or monetary reasons. Skågeby (2010, p.171) report that "the gift-giving literature highlights the division between the rationales of the market (utilitarian) and the rationales of personal relations (anti-utilitarian)". Nguyen and Munch (2011) also report dual motivations for romantic gift giving. They report "at least half of the population perceives gifting their romantic partners as more of an obligation and less of a pleasure" (p.117). Gifting for pleasure should be linked to higher perceived social gains, while gifting out of obligation should yield more cost conscious giving.

Based on this research, dual motives should also surface for gifting from online wedding registries. These dual motives yield one segment of gift givers who are primarily concerned with social benefits and the other with monetary costs.

Because an online wedding registry reports the price for each gift, a gift giver seeking a social benefit will largely ignore the price, provided the gift does not exceed his or her budget. This gift giver is likely to respond *positively* to more expensive gifts as they express more strongly and more clearly their social identity and affection. In contrast, a gift giver driven by monetary costs will react negatively to expensive gifts. This is because costs are more salient to their purchase decision. Thus, for both segments, the purchase likelihood is largely dependent on the gift's cost.

How can a gift giver identify high-priced gifts that appeal to the social benefit segment and low-priced gifts that appeal to the monetary cost segment? The marketing literature has firmly established that price judgments are a function of the actual price and its relationship relative to a reference price (e.g., Kahneman and Tversky, 1979). For example, listed retail prices for competitive brands can serve as an external reference price, which influences consumer purchase behaviour (e.g., Kumar et al., 1998; Lattin and Bucklin, 1989; Mazumdar et al., 2005; Moon et al., 2006; Winer, 1989). An external reference price is often estimated using average prices on the retail shelf. For a wedding registry, a comparable estimate is the average registered item price.

In contrast, an internal reference price is based on internal norms that can be formed from activities like a recent shopping experience. While an internal reference price also influences purchase behaviour, our focus will be on external reference prices, hereafter called reference prices.

Because online wedding registries post the prices for each item, a gift giver can quickly estimate a reference price. Once a reference price is established, gift givers in the social benefit segment should quickly eliminate gifts below the reference price, while those in the monetary cost segment should quickly eliminate gifts above the reference price. Therefore, gifts with prices above the reference price should appeal to the social benefit segment, while those below should appeal to the monetary cost segment. This yields:

H_1 In the social benefit segment, when the price of a gift is above the average registry gift price, the higher the price, the more likely the gift request will be fulfilled.

H_2 In the monetary cost segment, when the price of a gift is below the average registry gift price, the lower the price, the more likely the gift request will be fulfilled.

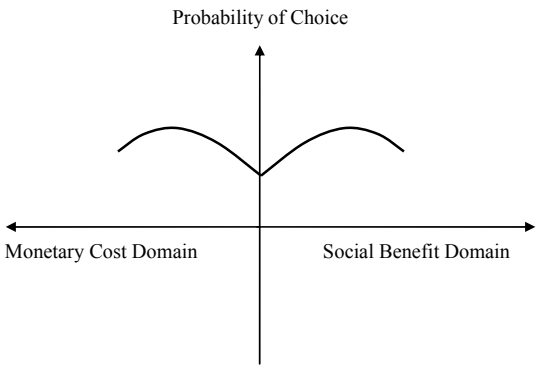
A key implication is that if the social benefit segment prefers higher-priced gifts (H_1) and the monetary cost segment prefers lower-priced gifts (H_2), the empirical relationship between a gift's price and the chance that it is fulfilled is V-shaped. This is because gifts at or near the reference price do not appeal to either segment. These gifts are 'stuck in the middle' and consequently have the lowest chance of being purchased.

Gift givers though may be averse to choosing a gift with an extremely high or an extremely low price. For the monetary cost segment, a very cheap gift can discourage purchase because it signals a lack of affection. It can also violate social norms in terms of "the givers perception of reciprocity and indebtedness to the receivers" [Nguyen and Munch, (2011), p.114]. For the social benefits segment, a very expensive gift can be outside the gift giver's budget. This yields:

H_3 In the social benefit and monetary cost segments, the effect of price on a gift's fulfilment probability increases at a decreasing rate and thereafter decreases. Thus, the curve follows an inverted-U shape in both segments.

Combining the three hypotheses, we expect the probability function of gift giving to be M-shaped, with an inverted-U shape in each segment (Figure 1). Note, the hypotheses focus on the probability that a gift is chosen. They do not address the *overall* fulfilment probability of a gift registry or the total gift giving *amount*. If the fulfilment probability function is M-shaped, as predicted above, an increase or decrease in price does not necessarily lead to a systematic change in a gift's fulfilment probability. Rather, the relationship between the reference price and the fulfilment probability is an empirical issue.

Figure 1 Illustration of the predicted reference price effect



3 Empirical analysis of wedding registry fulfilment outcomes

3.1 Data

Wedding registry and fulfilment data were downloaded from weddingchannel.com, a centralised wedding registry online website. It has links to the wedding registries of various retailers. The wedding registry data are randomly extracted from the website by querying the first two letters of recipients' first and last names. All of the weddings in our dataset took place during May 2005, a year prior to the data collection date to ensure all

gift purchasing activities had concluded. Therefore, our data capture a snapshot of wedding registry fulfilment outcomes.

The sample spans seven retailers, including four department stores (Bloomingdale's, Macy's, Kohl's, and Sears) and three specialty stores (Williams-Sonoma, Pottery Barn, and Crate and Barrel). The dataset contains complete lists of registered items, retail prices, product descriptions, requested quantities, and whether the item was purchased. We gathered a total of 555 registries as a calibration sample, including 32,764 registered items, of which 17,479 items had been purchased. Sixty-eight couples have wedding registries with more than one retailer, representing 150 registries. In Table 1, Panel A provides descriptive statistics for key registry characteristics across our sample of seven retailers. The average fulfilment rate is 53% in terms of items and 47% in dollar value.

The data analysis is limited to registries with fulfilment rates between 5% and 95%. This excludes registries with extremely high or low fulfilment rates, possibly due to limited choice or the wedding did not take place as planned. In Table 1, Panel B illustrates how fulfilment rates vary across various product categories. For example, dining and bar items have the highest item fulfilment rates (61%), while hardware has the lowest (20%).

Table 1 Descriptive statistics

A. Summary statistics of gift registries by retailers

Retailer	*No. of requested categories*		*Average requested price*		*Item price*		*Item fulfilment rate*	*Dollar fulfilment rate*
Department store								
Bloomingdale's	3.41	(1.22)	69.94	(35.93)	67.73	(9.90)	0.63	0.57
Macy's	3.91	(1.11)	47.74	(2.17)	47.11	(51.49)	0.47	0.41
Kohl's	4.18	(1.09)	19.11	(11.13)	16.29	(29.88)	0.45	0.40
Sears	3.56	(1.13)	51.62	(45.70)	43.02	(11.36)	0.22	0.15
Specialty store								
Williams-Sonoma	3.63	(0.97)	54.84	(36.14)	52.19	(69.30)	0.48	0.48
Pottery Barn	3.51	(1.03)	54.28	(41.77)	52.74	(123.42)	0.52	0.42
Crate and Barrel	4.13	(1.05)	21.69	(26.50)	18.85	(51.62)	0.63	0.55
Total	3.78	(1.11)	44.76	(36.29)	40.65	(75.19)	0.53	0.47

B. Summary statistics of gift registries by product categories

Category	*Registered items*	*Average price (std. dev.)*		*Item fulfilment rate*
Dining and bar	17,847	38.90	(56.02)	0.61
Kitchen	5,807	36.32	(61.57)	0.50
Home décor	1,928	28.18	(42.69)	0.42
Bed and bath	4,035	25.76	(39.74)	0.42
Appliances	1,073	94.71	(143.84)	0.56
Furniture	342	211.76	(356.93)	0.35
Hardware	464	37.01	(43.03)	0.20
Other	1,268	6.74	(111.77)	0.34
Total	32,764	40.65	(75.19)	0.53

Note: NStandard deviations are in parentheses.

3.2 Model specification

A three-level hierarchical logit model analyses the reference price effects on the fulfilment probability at the item-level, registry-level, and retailer-level. The modelling provides statistically efficient estimates, as well as more conservative standard errors and confidence intervals than the traditional regression models which ignore the presence of correlation within groups. In our model, the correlation within groups is at the registry-level and the retailer-level.

The fulfilment outcome (Υ_{jrs}) for gift j ($j = 1,\ldots,J_{rs}$) in registry r ($r = 1,\ldots,R_s$) and retailer s ($s = 1,\ldots,S$) follows a Bernoulli distribution. It is governed by the parameter ϕ_{jrs}, which is modelled as a logistic probability.

$$Y_{jrs} \mid \phi_{jrs} \sim Bernoulli\left(\phi_{jrs}\right), \text{ where } \phi_{jrs} = \frac{\exp\left(u_{jrs}\right)}{1+\exp\left(u_{jrs}\right)} \tag{1}$$

At the item-level, the attractiveness of a gift u_{jrs} is a function of social benefits ($sBene_{jrs}$) and monetary costs ($mCost_{jrs}$) as proposed in H_1 and H_2. Category-specific intercepts are also included, with other gifts being the baseline category. This yields:

$$u_{jrs} = \alpha_{0rs} + \alpha_{1r} sBene_{jrs} + \alpha_{2r} sBene_{jrs}^2 + \alpha_{3r} mCost_{jrs} + \alpha_{4r} mCost_{jrs}^2 + \kappa_j \tag{2}$$

Testing the effect of social benefits and monetary costs requires an empirical definition for an external reference price. Following earlier research on external reference prices (see, for example, Hardie et al., 1993; Kalwani and Yim, 1992; Kopalle and Lindsey-Mullikin, 2003; Mazumdar et al., 2005; Moon et al., 2006; Van Oest, 2013; Winer, 1989), we tested a variety of external prices such as the population average price (DIC – deviance information criterion: 35240.0), product category average prices (DIC: 34789.7), retailer average prices (DIC: 35113.3), and the registry's median price (DIC: 33850.2). The within-registry average price, which is used below, achieves the best fit (DIC: 33312.7).

Based on this external reference price, the social benefit is defined as the difference between the item price and average registry price when the item price exceeds the average registry price and 0 otherwise. Similarly, monetary cost is operationalised as the difference between the average registry price and the item's price when the average price exceeds the item's price and 0 otherwise. To ensure the social benefits and monetary costs are comparable across different registries; each one is normalised with respect to the maximum registry price ($MaxRprice_{rs}$). Thus, both values fall between 0 and 1. Specifically, each item is estimated as a social benefit $\left(\frac{Itemprice_{jrs} - AvgRprice_{rs}}{MaxRprice_{rs}}\right)^+$ or as a monetary cost $\left(\frac{AvgRprice_{rs} - Itemprice_{jrs}}{MaxRprice_{rs}}\right)^+$.

At the registry level, the random intercept α_{0rs} is modelled as a function of the number of categories ($Ncat_{rs}$) and a multiple-registry dummy variable, which equals 1 if a registry belongs to multiple registries; 0 otherwise. This yields:

$$\alpha_{0rs} = \gamma_{00s} + \gamma_{01} Mreg_{rs} + \gamma_{02} Ncat_{rs} + \gamma_{03} Ncat_{rs}^2 + \varsigma_{0rs}, \varsigma_{0rs} \sim N\left(0, \sigma_{\alpha_0}^2\right) \quad (3)$$

For *Ncat*, an inverted U-shaped relationship can arise because requesting too few categories can miss some popular gift categories, whereas requesting a very large variety can reduce the probability of an item's fulfilment. Multiple registries (*Mreg*) is also included as a control variable. This is because having more than one registry to choose from should reduce the chance that a specific item is fulfilled.

Finally, unobserved heterogeneity is modelled across retailers with retailer-specific intercepts ξ_s (using Sears as the baseline):

$$\gamma_{00s} = \mu + \xi_s + \tau_s, \tau_s \sim N\left(0, \sigma_s^2\right) \quad (4)$$

Overall, this three-level hierarchical model captures the item-level, registry-level, and retailer-level effects on gift fulfilment, while controlling for unobserved heterogeneity at all three levels.

3.3 Results

Table 2 reports the hierarchical Bayesian estimation results. Models 1 and 2 are benchmark models. Model 1 only has the control variables. Model 2 replicates the full model, but does not allow for unobserved heterogeneity. Model 3 uses the full model specification and corrects for unobserved heterogeneity. As expected, Model 3 has the best performance according to the DIC and is used to test the hypotheses. Statistical significance is based on two-tail tests.

Table 2 Item fulfilment estimation results (n = 32,764)

	Model 1		*Model 2*		*Model 3*	
	Baseline model (with heterogeneity)		*Full model (without heterogeneity)*		*Full model (with heterogeneity)*	
Intercept (μ)	–1.186	(0.286)***	–2.688	(0.450)***	–1.633	(0.449)***
Item-level						
Dining and bar	0.783	(0.078)***	0.727	(0.077)***	0.839	(0.091)***
Kitchen	0.158	(0.079)**	0.241	(0.076)***	0.174	(0.091)***
Home décor	0.167	(0.092)**	0.144	(0.093)**	0.196	(0.102)
Bed and bath	0.124	(0.084)	0.154	(0.081)**	0.014	(0.097)
Appliances	0.918	(0.104)***	0.889	(0.098)***	1.150	(0.119)***
Furniture	–0.839	(0.153)***	–0.347	(0.135)***	–0.678	(0.189)***
Hardware	0.011	(0.175)	–0.116	(0.174)	0.033	(0.194)
sBene [social benefit]			0.456	(0.263)**	0.974	(0.431)***
$sBene^2$			–0.926	(0.351)***	–2.880	(0.677)***
mCost [monetary cost]			2.782	(0.437)***	2.394	(0.719)***
$mCost^2$			–4.427	(1.350)***	–1.421	(0.623)***

Notes: Standard deviations are in parentheses. Statistical significance is based on two-tail tests with ** significant at the 5% level and *** significant at the 1% level.

Table 2 Item fulfilment estimation results (n = 32,764) (continued)

	Model 1 *Baseline model (with heterogeneity)*		*Model 2* *Full model (without heterogeneity)*		*Model 3* *Full model (with heterogeneity)*	
Registry-level						
Ncat [number of registered categories]			0.725	(0.185)***	0.231	(0.144)***
$Ncat^2$			–0.103	(0.023)***	–0.039	(0.019)***
Mreg [multiple-registry]			0.018	(0.018)	–0.141	(0.166)
Retailer-level						
Bloomingdale's	1.309	(0.263)***	1.327	(0.109)***	1.419	(0.344)***
Macy's	0.737	(0.278)***	0.702	(0.105)***	0.712	(0.398)
Kohl's	0.424	(0.388)	0.880	(0.102)***	0.548	(0.392)**
Williams-Sonoma	0.776	(0.277)***	0.886	(0.105)***	0.952	(0.379)***
Pottery Barn	0.906	(0.281)***	1.042	(0.105)***	1.113	(0.432)***
Crate and Barrel	1.515	(0.279)***	1.480	(0.104)***	1.690	(0.404)***
Dispersions						
$\sigma^2_{\alpha_0}[\alpha_{0rs}]$	1.168	(0.041)***			1.234	(0.050)***
$\sigma^2_{\alpha_1}[\alpha_{01s}]$					4.248	(0.399)***
$\sigma^2_{\alpha_2}[\alpha_{02s}]$					4.965	(1.183)***
$\sigma^2_3[\alpha_{03s}]$					12.090	(0.865)***
$\sigma^2_4[\alpha_{04s}]$					1.872	(3.099)***
$\sigma^2_s[\gamma_{00s}]$	0.307	(0.145)***			0.387	(0.174)***
DIC	35,575.7		42,847.3		33,312.7	

Notes: Standard deviations are in parentheses. Statistical significance is based on two-tail tests with ** significant at the 5% level and *** significant at the 1% level.

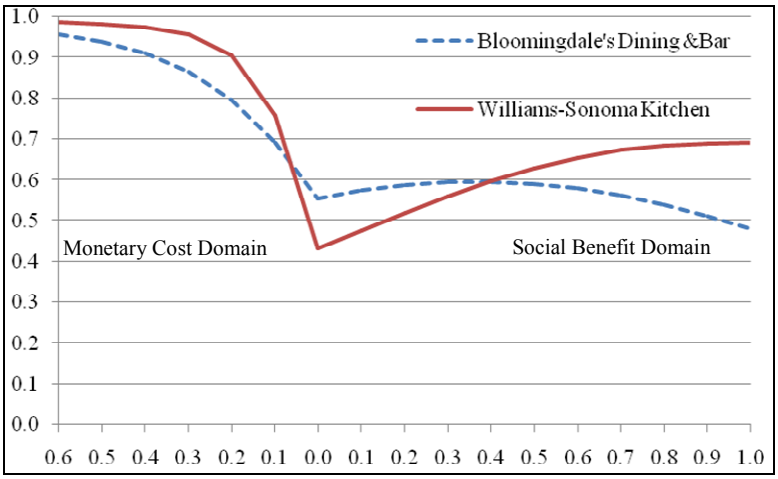
Consistent with H_1 and H_2, both social benefits and monetary costs have positive and significant signs (0.974, $p < 0.01$; 2.394, $p < 0.01$). Furthermore, the quadratic terms for social benefits as well as monetary costs have significant negative effects (–2.880, $p < 0.01$; –1.362, $p < 0.01$), which supports H_3.

It is interesting to note that two of the most popular categories are dining and bar items along with appliances. Consistent with the descriptive statistics, hardware and also furniture are among the least popular wedding gifts.

While the hypothesis testing results are statistically significant, it is also important to address their managerial significance. This is because with 32,764 items in each model, it is possible the large sample size is the main reason why these results are statistically significant.

Figure 2 illustrates the effects of social benefits and monetary costs by plotting the posterior means of the gift fulfilment probability curves. The results are for the dining and bar category at Bloomingdale's and Williams-Sonoma. These curves are based on the estimates from Model 3 in Table 2. All of the control variables use their mean values.

Figure 2 Asymmetric effects of reference prices (see online version for colours)



Note: In our dataset the maximum value of the monetary cost variable is 0.6, whereas the maximum value of the social benefit is 1.0.

For items in the social benefit domain, the curves reach the maximum fulfilment probability when *sBene* is around 0.4 for Bloomingdale's and 0.8 for Williams-Sonoma. In the monetary cost domain, the maximum for both curves occurs around 0.5. The graph also indicates overall higher fulfilment probabilities in the monetary cost domain. This may arise because lower-priced items are more affordable and are more likely to be combined into a multiple-item purchase. The height of the intercept is determined by the means of all the other control variables. Finally, the drop of the curve toward 1.0 in the social benefit domain likely results from the aversion to extreme price points (e.g., a $2,799 LCD TV is less likely to be purchased).

The Bloomingdale's data illustrate the influence of gift price changes on the fulfilment probabilities. For monetary costs, the predicted probability increases by 0.36 when an item's price drops from the average price (mean = $70) to the ideal price (i.e., 0.5, mean = $13.31) in the registry. This may not be music to retailers' ears, however, since the increase in the number of fulfilments appears to be more than offset by the decrease in purchase price. For example, Bloomingdale can expect a revenue of $31.5 (= $70 × 45%) from an average-priced item but only $13.2 (= $13.31 × 97%) from an item with the maximum fulfilment probability. This represents a 60% drop in revenue.

In contrast, in the social benefit domain, a higher price often leads to higher revenue. However, the predicted probability increases on average only by 0.05 in response to a price increase from the average (mean = $70) to the ideal point (i.e., 0.4, mean = $143.99).

In sum, Figure 2 illustrates the managerial significance of the two competing gift giving motives. The results show how selected items by gift givers appear to be primarily motivated by either

1 limiting monetary costs without offending the recipients

2 by maximising social benefits, without breaking the gift giver's budget.

A gift that is at or near the average registry price is not highly valued by either segment, so it has a relatively low fulfilment probability.

3.4 Limitations

Because of data limitations, some key variables, such as the gift giver's income (Garner and Wagner, 1991) and gender (Fischer and Arnold, 1990) are not included in our analysis. While income and gender have an important impact on gift giving, they do not offer an alternative explanation for the non-linear relationships estimated above.

A second limitation is that our data cannot identify when a gift giver combines multiple items into a single gift. Combining multiple items into a single gift registry may explain why low-priced gifts in Figure 2 have relatively high fulfilment rates. One of the authors, for example, received a $5 ice cream scoop as part of a multi-item wedding gift. Thus, while the empirical results support a bimodal relationship, the decline in fulfilment rates for very low-priced items is limited.

4 Discussion

In his reflections on consumer research in gift giving, Sherry (1996, p.223) wrote, "the registry phenomenon permits consumer researchers to perform a metasocial commentary of unparalleled significance. The creation of such a market, the institutionalisation of monadic giving, points to the transmogrification of culture that give rise to the original form. This ultimate commodification of the gift economy betokens fundamental culture change, which consumer researchers are in an advantageous position to explore".

As a first step toward understanding the effects of reference prices on online gift giving, our research finds that the pattern of fulfilment probability reflects the combined influence of a reference price and two competing motivations. These competing motives are a desire for social benefits as well as limiting monetary costs. One segment of gift givers places the highest priority on social benefits, while the second places the highest priority on limiting monetary costs.

In addition, both segments appear to avoid extremes prices in an online registry. This is probably because gift givers who want to limit their expenses want to avoid offending the recipients, while gift givers who want to maximise social benefits want to avoid breaking their budget. Because items that are at or near the average registry price do not appeal to either segment, they have relatively low fulfilment probabilities.

These results yield a number of important managerial implications for online retailers. First, a retailer's profit is a trade-off between an item's fulfilment probability and its profit margin. It does not always make economic sense to increase higher fulfilment rate at the expense of margins; this is especially true for items with lower than average prices.

Second, retailers can utilise information from existing databases to develop an intelligent recommendation system. For example, an intelligent recommendations system can make suggestions to the engaged couple if there are gaps in the wedding registry in terms of either item prices or excluded product categories. A smarter gift registry is a win-win for both customers and retailers because customers will feel less helpless – choosing gifts, even for oneself, can be a very daunting task – and retailers can expect higher fulfilment rates and more items in each registry.

Third, reference prices (and perhaps other contextual cues) are an important influence on choices in the registry context, especially for those gift givers who may be less certain about the preference of the recipient (e.g., casual acquaintances). When left alone, they may not be the most promising group of customers for the retailer since they may opt for the less expensive and presumably less profitable items. But retailers may get more return for their marketing dollars if they can steer those givers to higher-priced and more profitable items through contextual cues (reference price is one of them). For example, a retailer can easily provide information on the relative popularity of each item in a gift registry, which can motivate cost conscious givers to buy more expensive gifts.

In conclusion, our research offers empirical evidence for a better understanding of the motivational and contextual factors that influence online gift giving, but caveats exist. In particular, our analysis examines the outcome of gift giving in the registry context, but is mute on the process. Future research could address interesting research questions such as, does the reference price shift over time in such a dynamic environment? Does the dynamic nature of the process make it easier or more difficult for gift givers to make decisions? How does the option of combining multiple items in a single gift influence total spending? In short, online gift giving presents unique and significant marketing issues that deserve more research. A better understanding of this area not only contributes to the marketing literature but also offers practical recommendations to marketing practitioners to serve consumers better and to increase retailer profits.

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